Statistical hypothesis testing on the Cs-137 probability distribution, half-life of Ba-137m and characterization of the used Geiger counter

Paul Konrad and Richard Strunck

This experiment comprises the characterization of several radioactive specimen and the discussion of the respective measurement device for radioactivity, the Geiger counter. Latter included finding the threshold voltage $U_{\text{th}} = (510 \pm 6)\text{V}$, null effect ($n_a = (37.9 \pm 2.8)\text{min}^{-1}$) and sensitivity $Q = (0.204 \pm 0.018)\%$. In addition, the statistics of the radioactive decay of uranium trioxide were discussed. Via a χ^2-test it was shown that it can be described by a Poisson distribution. Finally, the half-life of Barium-137m was found to be$t = (154.1 \pm 1.2)\text{s}$.

Date of experiment: September 8, 2020
Submitted: September 11, 2020
Revised: September 16, 2020

1 Introduction

Since the end of the 19th century the phenomenon radioactivity is known. In the past century the areas of application have increased dramatically. Not only is it possible to gain energy from nuclear fission but also one can improve the healthcare system with the possibility of taking an X-Ray and optimising medical tests and treatment with the information gained. Nevertheless, nuclear fission can also be used in extremely life threatening ways e.g a nuclear bomb. The detonation is not the most dangerous impact but instead the radioactivity released by it. This is also relevant for nuclear accidents like the one that happened in Tschernobyl or Fukushima where the main damage is not caused by explosions but by the radioactivity contaminating the area. Therefore, it is important to analyze and categorize radioactive materials in order to estimate their impact on living creatures. In the following measurement radioactivity is discussed and the radioactive elements Cs-137 and Ba-137 are analysed as well as the Geiger counter, a tool to detect radioactivity.

2 Experiment

Radioactivity is a stochastic process that can be described by exponential decay. The number of decay events $-\mathrm{d}N$ expected to occur in a small interval of time $\mathrm{d}t$ is proportional to the number $N(t)$ of atoms present at time t. The decay rate of a one-decay process is described by the differential equation

$$\frac{\mathrm{d}N}{\mathrm{d}t} = -\lambda N(t) \tag{1}$$

$$= -A(t) \tag{2}$$

where $A(t)$ is the decay rate and λ is the decay constant. The differential equation is solved by:

$$N(t) = N_0 \mathrm{e}^{-\lambda t} \tag{3}$$

where N_0 is the value of N at time $t = 0$. This is called the decay law. The decay constant is related to the half-life $t_{1/2}$ of the radioactive element by $\lambda = \frac{\ln 2}{t_{1/2}}$.

2.1 Dose and dose equivalent rate of radioactive a specimen

The human body is permanently exposed to radioactive radiation. To estimate the radiation exposure the equivalent dose rate in the present laboratory was recorded over the course of the experiment in time intervals of 15 min. The equivalent dose rate and the equivalent dose absorbed in a time span of one year in the laboratory consisting of 8760 h as well as the equivalent dose absorbed during an estimated work time of 3.5 h are shown in Equation 4.

$$\begin{aligned} \dot{H}_0 &= (107 \pm 1)\frac{\mathrm{nSv}}{\mathrm{h}} \\ H_{3.5\,\mathrm{h}} &= (0.375 \pm 0.005)\mu\mathrm{Sv} \\ H_{1a} &= (0.94 \pm 0.01)\mathrm{mSv} \end{aligned} \tag{4}$$

This means even a supervisor staying in the laboratory for a whole year doesn't exceed the annual limit of 5 mSv and therefore should not suffer radiation toxicity. Using a dosimeter the dose rate of collimated γ-rays emitted from a Cs-137 sample was determined in a series of measurements. With a weighting factor of $Q = 1$ for γ-rays, the mean value of the dose rate $\dot{D}$ and it's standard error the equivalent dose rate $H = Q \cdot \dot{D}$ was calculated. The sample was stored in a lead cube with a sealable hole. The first measurement series was recorded directly above the unsealed hole, where the collimated γ-rays were emitted. For a time span of 4 h one experiences an equivalent dose of

$$H_{0\,\mathrm{m},4\,\mathrm{h}} = (26.95 \pm 0.55)\mu\mathrm{Sv}. \tag{5}$$

Additionally, measurements at a distance of about 1 m above the sample were taken. The same procedure as before was repeated and the background radiation Equation 4 was subtracted. The resulting equivalent dose is

$$H_{4\,\mathrm{h},1\,\mathrm{m}} = (0.61 \pm 0.13)\mu\mathrm{Sv}. \tag{6}$$

Due to the lack of a mounting the dosimeter had to be held by hand. This could have made the data imprecise since the dosimeter could have been misaligned to the maximum of the collimated γ-ray. Even at a distance of 1 m from the sample one could stay for $(33 \pm 7) \cdot 10^3$ hours to reach the annual limit of 5 mSv. It is still advised not to swallow the sample.

2.2 Activity of Caesium 137

The Caesium 137 specimen was delivered to the laboratories on August 21, 1978 with an activity of $A_0 = 3.70\,\mathrm{MBq}$. The half life of Cs-137 is about 30.2 [1] years. On the day of the experiment, September 8, 2020 the activity therefore is estimated to be

$$A = A_0 \mathrm{e}^{-\frac{\Delta t}{\tau}} = 1.41\,\mathrm{MBq} \tag{7}$$

where Δt is the time in months between the date of delivery and the date the experiments.

2.3 Probability distribution of a uranium trioxide coated tile

With a digital Geiger counter connected to a Personal Computer the radioactivity of a tile coated with uranium trioxide was observed. The radioactive specimen together with the Geiger counter tube were placed inside a lead cube to cover the experiment from outside noise and conversely the outside from the radioactive radiation. With this setup the number of decays in a predefined time span were counted and a measurement series was recorded.

2.3.1 Testing of Poisson distribution

Observing radioactivity is a sequence of testing for success (decay) or failure (no decay). This process is described by a binomial distribution. If, however, the number of trials n goes to infinity while the probability p of success tends to zero such that $np = \mu =$ const. the binomial distribution can be approximated by a Poisson distribution

$$P(k) = \frac{\mu^k}{k!}\mathrm{e}^{-\mu} \tag{8}$$

where k is the number of successes. Whether radioactive decay can be described by this is tested in the following.

With a gate time of $t = 1\,\mathrm{s}$ measurements were taken. The decay rates were counted and the result is shown in Figure 1. In total 6 classes occurred in the measurement. From the data the arithmetic mean and it's standard error were calculated to

$$\mu = 0.817 \pm 0.025 \tag{9}$$

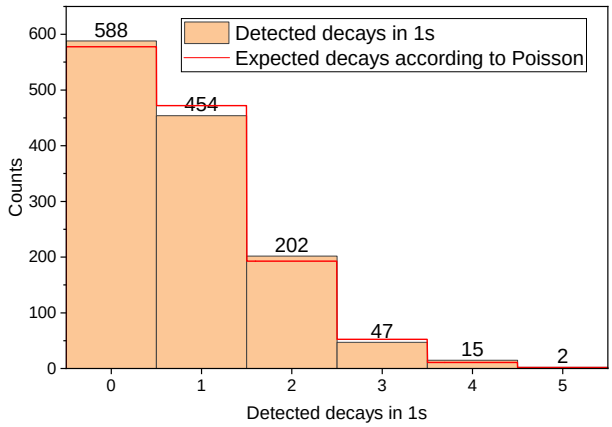

Fig. 1: Probability distribution of detected γ-quanta emitted from a uranium trioxide tile within a time interval of 1s. In red: the expected decays according to a Poisson distribution with mean 0.817 ± 0.025 (see Equation 9)

To test if the data follows a Poisson distribution a χ^2-test is performed. For the test to give valid results the classes were combined until their expected value was greater than 5. The combined classes together with their measured and expected occurrence and their χ_i^2-value are shown in Table 1. The sum of the χ_i^2 results in the total of

$$\chi^2_{1\,\mathrm{s}} = 3.3. \tag{10}$$

Since the number of measurements is fixed and the mean had to be calculated from the given data the total degrees of freedom for the χ^2 distribution is 3. The null hypothesis is that the data is Poisson distributed. With a significance level of 5 % the corresponding threshold value is $\chi^2 = 7.81$. Therefore, the null hypothesis is accepted.

2.3.2 Testing of Gaussian distribution

For a sufficiently large mean value μ the Poisson distribution can be approximated by a Gaussian one. To increase the mean value of detected events for the given sample the gate time was increased to $t = 10\,\mathrm{s}$ and like in subsubsection 2.3.1 the decay rates were counted. The results are shown in Figure 2. The mean with it's standard error and the standard deviation are

$$\mu = 7.928 \pm 0.090 \tag{11}$$

$$\sigma = 2.9. \tag{12}$$

After combining classes until their expected value was greater than 5 a total of 15 classes with their measured and expected occurrence and their χ^2

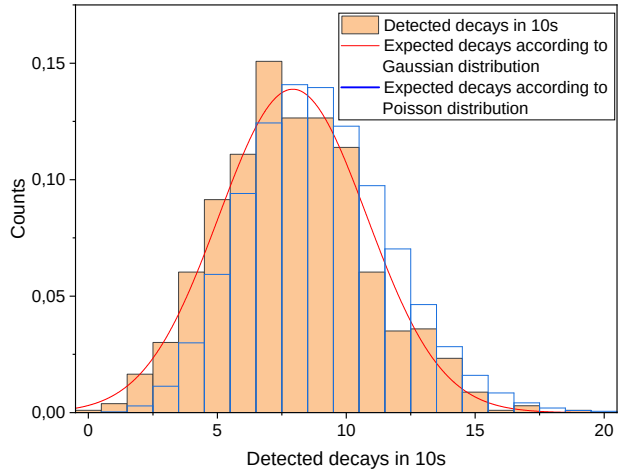

Fig. 2: Probability distribution of detected γ-quanta emitted from a uranium trioxide tile within a time interval of 10 s. In red: The expected decays according to a Gaussian distribution with mean 7.928 ± 0.090 and standard deviation 2.9 (see Equation 12

value are shown in Table 1. Since the mean value and it's standard deviation were calculated with the data and the number of measurements is fixed a total of 13 degrees of freedom are used for the χ^2 distribution.
For the present data one obtains

$$\chi^2 = 53.42. \tag{13}$$

Values greater or equal to (13) occur in only 7.6×10^{-5} % of cases . With a 5 % probability of error the corresponding threshold value is $\chi^2 = 22.36$. Therefore the null hypothesis that the data is not Gaussian distributed is accepted on a significance level of 5 %. To test for a Gaussian distribution the data is plotted in Figure 3.

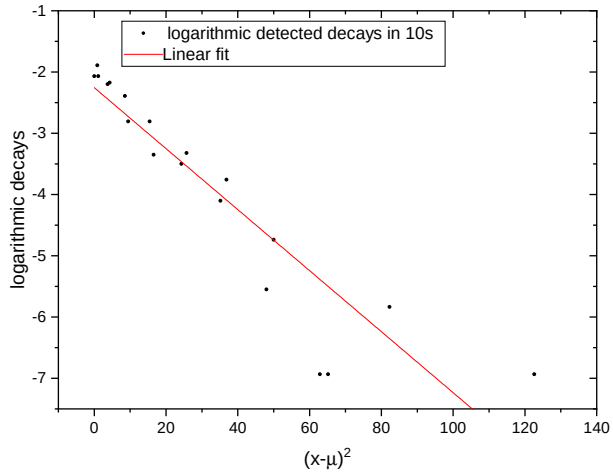

Fig. 3: The logarithm of the class occurrence plotted over the difference between class value and mean value (12) $(x-\mu)^2$. A linear fit is applied to test for a Gaussian distribution.

The counted events are plotted logarithmically

over $(x-\mu)^2$ where x is the class value. If the data follows a Gaussian distribution one obtains the standard deviation by a linear fit with slope m. The expected value can be calculated considering that the data follows a Poisson distribution. The two calculated values are

$$\sigma = \frac{1}{\sqrt{2m}} = 3.17 \pm 0.16 \tag{14}$$

$$\sigma_{\text{Poisson}} = \sqrt{\mu} = 2.816 \pm 0.016. \tag{15}$$

The values are not compatible within range of their error. This also supports the hypothesis that the data is not Gaussian distributed. This shows that even with a timespan of 10 s the mean value isn't large enough to approximate the distribution with a Gaussian distribution.

2.4 Counter tube characteristics

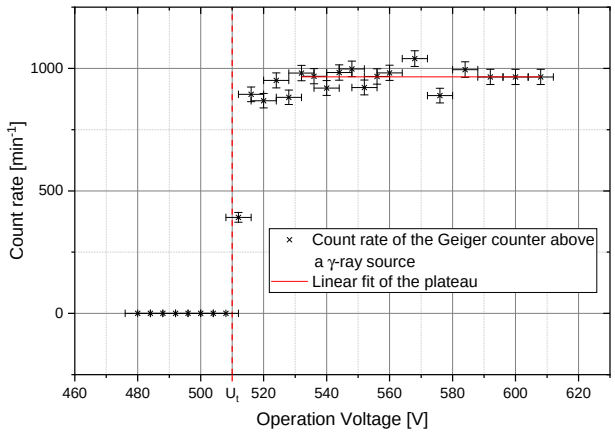

Fig. 4: The plateau voltage is denoted by the red linear fit.

In this part of the experiment the characteristics of a Geiger counter are discussed. To that end the Geiger counter is placed above the opening of the radioactive Cs-137 used in subsection 2.1 that is stored in a lead cube. Starting at ≈ 480 V the pulse rate is measured up to voltage of 608 V. This is shown in Figure 4; the operation voltage has to be increased until it reaches the so called plateau voltage. The Geiger counter only shows a response to incoming radiation above the threshold voltage of

$$U_{\text{th}} = (510 \pm 6)\text{V}. \tag{16}$$

By increasing the voltage the saturation is reached and the detected count rate is not depending on the operation voltage. In this voltage range incoming radiation quanta can set off electron avalanches, which are cause to the measured ionisation current of the Geiger counter.

2.4.1 Null effect of counter tube

The Null effect of the Geiger counter was measured. At an operation voltage of ≈ 580 V the counter rate was recorded for 5 minutes without any radioactive specimen actively measured. Still, on average about every two seconds a radioactive quantum was detected ($n_s = (37.9 \pm 2.8)\text{min}^{-1}$). Then, the Null effect was taken from inside lead block, which should block most radiation. With this setup only one in every four seconds on average a radioactive quantum was detected ($n_s = (15.6 \pm 1.3)\text{min}^{-1}$). The difference between the two locations' pulse rate is $\Delta_n = (22.3 \pm 1.2)\text{min}^{-1}$. There may be several interpretations of this phenomenon. On the one hand, the radioactive measurements might be due to atmospheric background radiation e.g. caused by gaseous Radon in the air. On the other hand, the radioactive specimen from the laboratory might be the cause to this effect. The lead cover was not entirely closed due to an opening for the cable of the counter tube. A combination of both could explain that in the lead shrouded case radiation was still detected but at a much lower rate.

2.4.2 Half-life of Ba-137m

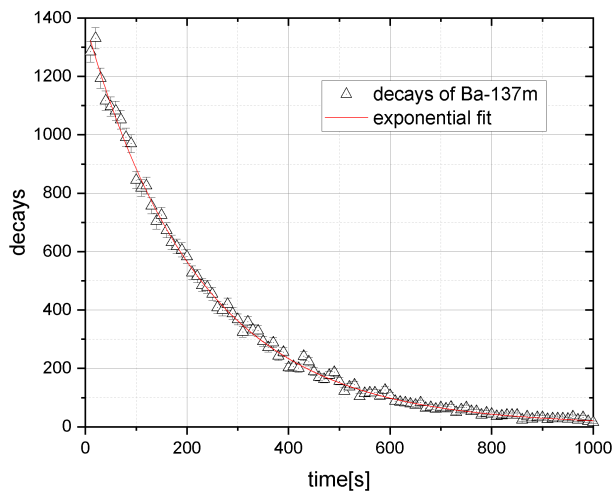

Fig. 5: The decay rate of a Ba-137m sample over time. The reader notes the exponential functionality behind the Ba-137m decay.

The element Cs-137 decays either into Ba-137 or in the excited state Ba-137m which itself is ra-

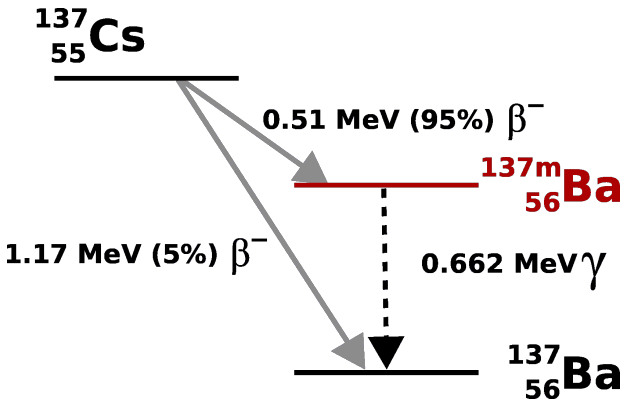

Fig. 6: The decay scheme of Caesium-137 [2]

dioactive and decays under emitting a γ-Photon (see Figure 6). One can split Ba-137m from Cs-137 through hydrochloric acid mixed with a bit of sodium chlorid. The gained sample of Ba-137m was observed with the counter tube and it's half life was determined. The decay is shown in Figure 5. The measurement was recorded digitally by a Leybold-Digital-Counter and a Personal Computer. The half-life was determined from the exponential fit data to $t = (154.1 \pm 1.2)$s. This measurement is compatible with the literature value of $t = 153.1\,\mathrm{s}$ [3] within the range of its error.

2.4.3 Counter tube response characteristic to γ- quanta

The response characteristic of the LEYBOLD-Digital-Counter was determined. Intuitively, it is defined as the quotient of the γ-quanta registered by the Geiger counter and the γ-quanta hitting the Geiger-counter. The specimen was placed at a distance of $d = (12.5 \pm 0.5)$cm from the Geiger counter tube. The detector area was given in the laboratory instructions as $a = 11\,\mathrm{cm}^2$. The expected detected activity is

$$A_{\text{expected}} = \frac{a}{4\pi d^2} A = (7.90 \pm 0.63)\text{kBq} \quad (17)$$

where A is the current activity derived in Equation 7. The real detected activity is $A_{detected} = (16.08 \pm 0.17)$Bq. The sensitivity therefore is

$$Q = \frac{A_{\text{detected}}}{A_{\text{expected}}} = (0.204 \pm 0.018)\%. \quad (18)$$

Since the specimen wasn't taken out of the lead cube it was stored in and the γ-quanta were only allowed to escape the lead cube through a tiny hole the real responsiveness of the counter tube is probably bigger than the value obtained.

3 Summary

In this experiment three radioactive specimen were probed. The activity of the used Cs-137 sample was determined, the distribution of the decay of uranium trioxide was tested and the half-life of Ba-137m was determined. It was found that over a measurement with about 1000 samples taken in time intervals of 1 s the detected decay of uranium trioxide can be described by a Poisson distribution with a significance level of 5%. Conversely, one showed that the same specimen, observed with time intervals of 10 s, is not Gauss distributed on a 5% significance level. Following that, the characteristics of a Geiger counter were determined. It was shown that the Geiger counter starts the effective measurement process at a threshold voltage of $U_{\text{th}} = (510 \pm 6)$V. Also, the null effect - defined as the radiation that the Geiger counter indicates even if no radioactive specimen is placed close to it - is found to be ($n_a = (37.9 \pm 2.8)\text{min}^{-1}$). However, if the counter tube is covered by lead this value deviated by $\Delta_n = (22.3 \pm 1.2)\text{min}^{-1}$ leading to the conclusion, that the lead shield very effectively reduces the null effect . This consideration was enhanced by the discussion of how harming the exposure to the radioactivity of the laboratory might be for a experimenter over the year, which came out to be tolerable within the 5 mGy per year. The sensitivity of the Geiger counter was found to be $Q = (0.204 \pm 0.018)\%$. Finally, the half-life of Ba-137m was determined as $t = (154.1 \pm 1.2)$s.

4 Appendix

class	H_{exp}	H_{th}	$\frac{(H_{exp}-H_{th})^2}{H_{th}}$
0	588	578	0,185276297
1	454	472	0,694289823
2	202	193	0,427319174
3	47	53	0,587469064
≥ 4	17	13	1,406029678

Table 1: The data for a χ^2-test of Poisson distribution

class	H_{exp}	H_{th}	$\frac{(H_{exp}-H_{th})^2}{H_{th}}$
0	1	5	3.39070472
1	4	12	5.27032824
2	17	24	2.18338176
3	31	44	3.75991189
4	62	70	0.95909174
5	94	100	0.32906553
6	114	126	1.08213425
7	155	140	1.50762828
8	130	139	0.61339479
9	130	122	0.4655609
10	117	96	4.83180988
11	62	66	0.2531354
12	36	41	0.51319259
13	37	22	10.0780239
14	24	11	16.6814979
≥ 15	14	10	1.5035388

Table 2: The data for a χ^2-test of Gauss distribution

References

[1] `https://www.nist.gov/pml/radionuclide-half-life-measurements/radionuclide-half-life-measurements-data` last visited September 11th, 2020

[2] `https://commons.wikimedia.org/wiki/File:Caesium-137_Decay_Scheme-de.svg`, last visited September 10th,2020

[3] `http://www.periodensystem-online.de/index.php?id=isotope&el=56&mz=137&nrg=0.6617&show=nuklid`, last visited September 9th, 2020